BT102
PAST PAPERS
FINAL TERMS

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Incidence and Prevalence 2 marks

Incidence: # of new cases in a given time period in a population

Prevalence: New + existing cases

PRIMARY AND SECONDARY IMMUNE RESPONSE

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making

antibodies. ***Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are***

secreted and then 7-10 days after, the same B cell starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen

exposure and is called a primary immune response. It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of

antibodies that are achieved in a very short time (within 2-7 days). This is called a secondary immune response. It is also

called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second

exposure by the same antigen (to which the immune system has already been exposed). The accompanying Blue line in the figure depicts a primary and a secondary response.

Adaptive Immune Response can be acquired naturally or artificially. Each of these categories can further be divided either into active or passive type.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a

microbe but then recovers from that illness making antibodies.

Naturally Acquired Passive Immunity: A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.

Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes

antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.

Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person

from an illness or a snake bite, this constitutes an artificially acquired passive immunity type. The following figure illustrates these four types.

Vegetative Hyphae and Aerial hyphae 2 marks

Vegetative Hyphae: Hyphae that are embedded in the medium and are used to obtain food are called vegetative hyphae.

Aerial or Reproductive Hyphae: The portion of hyphae that is concerned with reproduction. This portion is projected

. Define and type of algae. 5 marks

Define :

The definition of algae is a single or multi-cellular organism that has no roots, stems or leaves and is often found in water

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such

as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly

photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth.

Types of algae:

1. **Phaeophyta (Kelp)**: Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell walls, are multicellular and contain chlorophyll a and c, xanthophylls. These store carbohydrates and are used

for algin which is a thickener in ice creams. Can grow upto 20 cm in one day.

2. **Rhodophyta: Red Alage,** these are reddish in color and contain cellulose in their cell walls. Most are

multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan (both are used as thickeners of foods).

3. **Chlorophyta: Green Algae:** have all the features of red algae except that chlorophyll is of a and b types only.

These algae are believed to have given rise to terrestrial plants.

4. **Bacillariophyta**: Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and

c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated

by eating contaminated mussels. Diatoms: Come in beautiful colors and shapes.

5. **Dinoflagellata:** These are unicellular algae collectively called as plankton (free floating organisms). These cells

have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll a and c,

and carotene. These algae store starch in them. Dinoflagellates in the genus Alexandrium produce neurotoxins

(called saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and clams. Blooms of Alexndrium gives ocean a deep red color commonly known as red tide.

6. **Oomycota (Water molds):** They are basically decomposers of dead matter. Superficially, these resemble with

fungi; however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores

are called zoospores. Terrestrial members of water molds are also plant parasite. Phytophthora infestans was

responsible for Irish potato blight during mid-1800s. P. ramorum causes sudden oak death.

HYPERSENSITIVITY AND ITS TYPE 5 marks **What is hypersensitivity?**

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way

for which it has made to. So when an antigen gets into the body, immune system recognizes it as foreign and

start making antibodies against the antigen. However, the consequences are bad and damaging to the body.

It may be called an abnormal antigenic response.

Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens

Types of hypersensitivity:

Hypersensitivities can be grouped into 4 types based on the types of antibodies or cells involved.

1. **Type I (Anaphylactic)**: Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated.

2. **Type II (Cytotoxic):** IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement.

3. **Type III (Immune complex mediated):** Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.

4. **Type IV (delayed type cell-mediated)**: CD4+ helper cells along with macrophages are involved.

. **Define antigen and epitope**

1. Antigen: A foreign substance that provokes immunity is called an antigen.

The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease.

More specifically called as an immunogen. So immunogen and antigen although slightly different but used

interchangeably. o Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens.

Carbohydrates and lipid alone are poor antigens. o Antigens are foreign to the body. o Degradability: Inert molecules are not antigenic.

2. Epitopes: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those act

as a small antigen. These are called epitopes.These epitopes are also called as antigenic determinants. o Most

antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

A small molecular region of an antigen that binds to a particular antibody or antigen receptor on a T cell; an antigenic determinant. A single antigen can have multiple epitopes

4. Describe two transported system in the blood and why it is necessary

Cardiovascular and lymphatics are the two transport systems of the body by which cells of the immune system patrol the whole body.

Cardiovascular system consists of the heart and associated blood vessels (arteries and veins) in which blood circulates

lymphatics are channels that start from tissue spaces and collect tissue fluid (called lymph) from tissue spaces and bring

them back to the heart. Along these lymphatic channels, there are strategically placed lymph nodes which filter lymph

for microbes much like blood is filtered in the spleen. Lymph nodes contain aggregations of B and T cells along with macrophages and dendritic cells which interact with each other inducing adaptive immune responses.

Why Cardiovascular and lymphatics are needed?

Well, in unicellular organisms, nutrients can be absorbed directly from the environment in which a unicellular organism

is living. However, a multicellular organism does not have this easy access to the nutrients. So, nutrients must surround

the cells in a multicellular organism. These nutrients are provided by cardiovascular system which pumps blood to all

organs. Remember that only fluid part of blood leaves blood vessels. RBCs never leave blood vessels. Sometimes

proteins also leak from these blood vessels; however, once they leak from the blood vessels, they never can enter back

into blood vessels because of their bigger size. If proteins remain in the tissue spaces, they exert osmotic pressure which

attracts water into these space resulting in edema. However, nature has provided another mechanism in the form of

lymphatic channels which start from tissue spaces and can drain tissue fluid and proteins back to the heart. This is the role of blood vessels and lympahtics that they ensure

circulation of nutrients throughout the body without causing any edema or abnormality. B cells and T cells also keep

circulating between these two channels (cardiovascular and lymphatics). Unfortunately, microbes can also use these two

systems to spread themselves in the body. That is the reason, spleen and lymph nodes are placed in these channels by

the nature to deal with these microbes that find their ways into blood and lymph. Lymphatic System consists of the

following: o Bone marrow o Lymph nodes o Spleen o Tonsils o Thymus o Mucosa associated o Bursa of Fabricius in birds

5. Disk diffusion method

A standardized inoculum of the organisms is spread on agar plate. Filter paper disks soaked with anti bacterials are then

placed on the lawn of the organism. Plates are incubated at 37oC and results are recorded. Drugs diffuse from the paper

disks radially and if the organisms are sensitive to the drug, a clear area with no growth around the disks develops. The

size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for

recording the results of sensitivity test. Results are reported as sensitive or resistant or intermediate (between the two

extremes).

6. **How microbes develop resistance in antimicrobial?** See current GDB for answer

7. **Define immunity and difference between innate immunity and adaptive immunity.** 10 marks

What is immunity?

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system

against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli,

Staph or Salmonella at the same time (if these pathogens are present there).

Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they

prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes.

So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells

phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive Immunity: This will be studied in detail in a separate lecture soon.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the

body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened

immune response against E. coli because it has memory cells developed in the body which upon second stimulation

proliferate quickly and mount a greater defensive response.

O Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes)

Cellular immune response (T cells are made which kills microbial harboring cells).

8. Agglutination and Neutralization 2 marks

Agglutination: Antibodies can agglutinate antigens which are then cleared as a group by macrophages

Neutralization: Toxins bind with antibodies and they become incapable of binding to their target receptors thus the toxins become harmless.

9. What is plaque 2 marks

Plaques are equivalent to bacterial colonies. Basically a plaque is an area where the virus infection has lysed the cells.

These plaques are formed on cultured cells in a culture dish. A monolayer of the cells is infected with the virus to be

enumerated and then the culture is overlaid with the semisolid agar so that infected and lysed cells do not move away

from the site in the dish. Plaques can be counted easily much like bacterial colonies

10. Difference between capsule and slime

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

Slime: If glycocalyx is loosely attached to the cell, it is called slime.

11.PCR method 3 marks

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of a denaturing stage (about 95oC), primer

annealing (50 to 56oC) and extending stage (72oC). A primer is a fragment of DNA which is specific to the signature

sequence. Two primers span the signature sequence actually. At the end of 30 or so such cycles, enough DNA

accumulates that one can visualize on an agarose gel easily.

12. Type of T cell T helper

cells

TH1 (CD4+)

TH2 (CD4+)

T cytotoxic cells (CD8+)

13. Inhibits growth cell wall drug name

Penicillins and cephalosporins are the major antibiotics that inhibit bacterial cell wall synthesis. They are

called beta-lactams because of the unusual 4-member ring that is common to all their members.

Microbiota factor and distribution

composition Nutrients:

Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2. Host defenses, level

of stress and age of an individual also matters for allowing microbiota of an individual.

15. Capsule function 2 marks

The capsule impairs phagocytosis and help the organisms evade the immune system; however, antibodies can help cells

of the immune system in phagocytosis of such encapsulated organisms. Only a few bacteria produce capsule which include Strep pneumonia, Klbesiella pneumonia.

16. Viral replication stages

Viral multiplication goes through various stages in the cell. These include viral attachment, entry into the cell by

pinocytosis or fusion with the plasma membrane, uncoating of the virus, biosynthesis of capsids and nucleic acids,

maturation (assembly of the virus) and final release of the virus from the cell.

17. Economic benefits of fungi

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively).

However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 (some

ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is

also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger;

cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples

of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

18.Types of symbiosis

Interaction of microbiota with the host leads to a relationship called symbiosis. There are a few outcomes of this

symbiosis which we must know. The two participants of this relationship are microbiota and the host. At least one of

these participants is benefited and sometimes both can be benefited from this mutual relationship. Different names or

terms have been given to these mutual relationships as under:

Mutualism: both are benefited Parasitism: one is harmed.

Commensalism: One is benefited, other is unharmed.

19. Inflammation and write its consequence 5 marks

Damage to body tissues triggers a local defensive response which is called inflammation. It causes redness, pain, swelling, heat, and loss of function of that organ

During inflammation, various mediators of inflammation are released, some of which include:

Histamine

Kinins: Almost all cells.

Prostaglandins by damaged cells

Leukotrienes by mast cells

Cytokines are released (TNF-alpha), and they cause release of acute phase proteins from the liver

20. Cause and diagnose of anaphylaxis

The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic.

- ***Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.***

Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific

IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation.

Moreover, IgE levels can be measured by ELISA. Measurement of IgE is more practical as nature of allergens is not known most of the time.

o Rx: Epinephrine & antihistaminic drugs provide a means of treatment.

21. Two name of bacteria which is capsulated

Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

22. What is Epidemiology and its types

The study of where and when diseases occur and how they are transmitted in a population.

There are three basic types of investigations relating to epidemiology

1. ***Descriptive Epidemiology***

2. ***Analytical Epidemiology***

3. ***Experimental Epidemiology:***

Target site of antibiotics

Capsomere

Capsid is made up of protein subunits called capsomeres that are arranged in 3 possible ways to give a helical, or icosahedral or complex symmetry to the viruses.

Capsomeres: Subunit called protomers aggregate to form capsomeres.

23. Character of antigen

Antigen: A foreign substance that provokes immunity is called an antigen.

More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably.

Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens. ***Antigens are foreign to the body.***

Degradability: Inert molecules are not antigenic.

24. Algae structure

Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes and leaflike blades. There is no vascular tissue in these algae. Also, the stipe is not lignified or

woody, so it does not provide support to the weed. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float

. Acid fast stanning 5 marks

Some bacteria such as Mycobacteria have a waxy material in their cell wall.

Specific name for this waxy substance is mycolic acid

Mycobacterium tuberculosis: Causative agent for tuberculosis

M. leprae: Causative agent for leprosy

- Principle of Acid-fast staining: Mycobacteria are lipophilic, not easy to stain

But once stained, they are resistant to acidic alcohol decolorization process.

- **<u>Overall Procedure for Acid-fast staining:</u>**

Carbol fuchsin: It is a primary stain Heat is used to enhance penetration of carbol fuchsin.

Acid alcohol is a decolorizer.

Methylene blue is a counterstain.

Please note, Mycobacteria are gram positive, however, they look red in acidfast staining.

28. Capsule staining 10 marks

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

– Capsulated organism make glistening colonies as seen in the accompanying diagram below:

Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

Thickness of the capsule depends upon the culture conditions.

Capsules are mostly are water soluble. Also remember that capsular organisms usually make the broth viscous and stringy

Capsule staining procedure

1. ***Prepare a thick smear*** in a loopful of congo red (1%) stain
2. ***Fixation in acid alcohol for 15 sec***
3. ***Wash with dH2O***
4. ***Cover it with acid fuchsin for 1 min.***
5. ***Wash with water***
6. A bacterium stains red, capsule remains colorless and is seen in the dark blue background as seen in the accompanying diagram.

Define vaccine and its types 10 marks

What is vaccine:

A product that creates immunity against an infection is called a vaccine. It could be the whole organism or a part of the organism.

The observation that people or animals once recover from an infection do not get sick easily to that organism again led to the discovery of vaccines and vaccination.

Types of vaccine:

Live Attenuated Vaccines: These are weakened organisms so that they do not harm the body, but divide or replicate

in the same way as pathogenic version of the organism does in the body. So, when a live virus or bacteria multiplies in

the body, it stimulates the immune system to create an adaptive immune response which is protective against the disease for which vaccine was given.

The benefits of live vaccines are a better immunity because live microbes mimic actual infection (although they do not

cause harm to the body). Live vaccines induce both humoral and cellular immune responses together, so live vaccinesa better choice if available. Because live microbes multiply in the body, this replication results in the increase of

antigenic amount needed for stimulation of the immune system. Also, this antigen remains for a long time in the body,

continuous presence of antigen continuously keeps the immune system active for longer time resulting in long term immunity.

Inactivated Killed Vaccines: Sometimes a nonpathogenic version of the microbe is not available, and to make a

vaccine in this situation, the organism is killed with formalin or phenol and then used as a killed inactivated vaccine.

Because organisms in such a vaccine do not replicate in the body, frequent boosters are required to achieve a good level

of immunity (antibodies). Also, remember that this is a killed vaccine, so classical cellular immunity is not created with killed vaccines.

Toxoids: Some diseases are caused by bacterial toxins, and not by bacteria. In such cases, ***toxin is modified***

(inactivated) and used as a vaccine. Such a vaccine is known as a toxoid. Tetanus toxoid is a good example.

Conjugated Vaccines: Sometimes special parts of organisms are required for generating protective immunity.

However, those substances are poor antigens if used alone. In order to enhance immunity, such poor antigens are

attached to proteins, and such vaccines are called conjugated vaccines. ***Haemophilus influenzae type b is a good example.***

DNA vaccines: We now know that sometimes only certain proteins are required for generating a good immune

response. In such cases, the genes responsible for those proteins can act as vaccines. DNA coding for such a protein is

injected into cells, and that DNA gets transcribed by the cells of the body into the required protein and that protein

induces the antibody production (because that protein is foreign to the body). ***West Nile Virus is a good example of DNA vaccine***

Subunit vaccines: Also called ***recombinant vaccines,*** such vaccines are composed of a portion of viral or bacterial

proteins that can induce immunity. In other words, this is another version of DNA vaccines. In subunit vaccines, antigenic

epitopes are included in the genome which is inserted into an expression vector that makes those subunit proteins.

These subunit proteins are collected and used as vaccines to stimulate the immune system. Again, these act like a killed

vaccine. There is another mechanism being explored in these days for recombinant vaccines. A subunit vaccine for Foot and Mouth disease virus is available.

Such genes are being incorporated in plants now that are acting like an expression vector. Animals can eat those plants

and subunit vaccines can induce immune response through M cells of the intestine.
Success is limited though so far.

. Describe features of fungi 5 marks

These are simple eukaryotic cells. Some are ***unicellular, others are multicellular (thallus);*** however, they ***lack tissues such***

as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are

mostly photoautotrophs; ***however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the***

face of the earth. Microscopic exam is needed to identify unicellular and filamentous algae. However, multicellular algae

that are commonly known as seaweeds are macroscopic in nature and can be identified morphologically without the

help of a microscope. ***Four groups of such algae include blue-green algae, green algae, brown algae and red algae. These*** algae are located in the sea at various locations and absorb light of various wavelengths, hence red algae are located r

from the surface and can use blue light from the sun as blue light is of shorter wavelength and can penetrate deep in the

sea. Please see the accompanying figure (above) for a better understanding of how wavelength relates with various

algae. Also remember, blue-green algae need magnification in order to be correctly identified, although they are not microscopic.

31. Treatment of hypersensitivity ii 3 marks

This type of hypersensitivity can be diagnosed by history and symptoms. Glomerulonephritis and cellulitis is very

common. Deposition of immune complexes in the kidneys can be diagnosed by presence of IgG in kidney cells by using anti-IgG antibodies. 3

What are antibiotics 3 marks

antibiotics are used to kill organisms in disease states, some antibiotics are not very effective for this purpose; however,

these antibiotics could be used in food to prevent food spoilage. Nisin and natamycin prevent spoilage of cheese

33. Define virion and viroid 2 marks

A virion is an infectious particle composed of nucleic acid that is surrounded by a protein coat (capsid). Viruses are

classified based on their nucleic acid and protein coat. Viruses have an extracellular and an intracellular state. ***Outside of a cell, in the extracellular state, a virus is called a virion.***

VIROIDS: They are composed only of circular single stranded (ss) RNA.

34. Write entry point of microorganism 2 marks

Mucus membranes: These include linings of digestive tract, respiratory tract, urogenital system, and conjunctiva

Skin: Surface area wise, skin is the largest organ of the body. Unbroken skin is impermeable to organisms Parenteral Route: Injury to the skin or mucous membrane leads to easy access for the organisms in the body. Any cut

received on the skin for example becomes a portal of entry

35. Role of thin clonal selection 2 marks

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a B

cell encounters its specific antigen in the body, it starts proliferating into a clone which ultimately becomes plasma cells

and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store

and select a shoe size that fits you very well. Note, that the shoe that you selected was made randomly without any regard to your foot size. **35.Antigen presenting cell 3 marks**

These cells are able to present antigens to B cells for making antibodies by B cells.

There are three cells that fall into this category:

Dendritic cells, macrophages, and B cells

36.How antibodies provide defense against microbes 5 marks

37.Reverse transcription 2 marks

A reverse transcriptase is an enzyme used to generate complementary DNA from an RNA template, a

process termed reverse transcription

38.Bactericidal and bacteriostatic drug 2 marks

Bactericidal: Those antibiotics that kill and lyse the cells are called bactericidal.

Bacteriostatic: Such antibiotics stop the growth of the organisms, they do not kill or lyse the cells. Cells of the immune system then clear them from the body

. Define plasmogamy karyogamy and meiosis 3 marks

1.***Plasmogamy***: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-). Both cells are haploid (one set of chromosomes)

2. ***Karyogamy The (+) and (-)*** nuclei fuse to form a diploid zygote nucleus.

3. ***Meiosis***: The diploid nucleus gives rise to haploid nuclei (sexual spores).

. Mechanism of hypersensitivity iv 5 marks

It is called as delayed type because its bad effect becomes visible 2-3 days after exposure to the antigen. This is the only

hypersensitivity that involves cells rather than antibodies. Robert Koch discovered this when he injected tuberculin in

patients. Contact dermatitis is the most common example of this kind of hypersensitivity. This is basically an interaction

between an antigen, T cells and macrophages. Transplant rejection is another good example of delayed type IV

hypersensitivity. Reaction to hair dyes is another example. Poison Ivy is another example in which contact with this plant

initiate type IV hypersensitivity response. Chemicals from poison ivy interact with skin proteins and change them into

foreign-looking protein molecules. Body's immune system mounts an immune response against these proteins resulting in a damaging

type of intense inflammatory response 41. **Cytokines 2 marks**

Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions. Cytokines recruits more

immune cells and also stimulate adaptive immune response

42. **Note on infectious disease tr**

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include

blood and other body fluids, drugs, and intravenous fluids.

A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. o Arthropods are vectors

for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.

Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

1. **Karyogamy plasmogamy and meiosis? 2MARKS**

Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-)

Both cells are haploid (one set of chromosomes).

Karyogamy: The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

Meiosis: The diploid nucleus gives rise to haploid nuclei (sexual spores).

2. **Advantage of Dry Heat relatively? 2 MARKS**

Advantages of dry heat sterilization include: it is inexpensive, it does not pollute or cause toxic fumes, and it does not corrode or rust metal objects.

Disadvantages include: it is relatively slow and many objects cannot withstand the high temperatures.

3. **How Microorganism cause damaged the body?**

Infection with a pathogen does not necessarily lead to disease. Infection occurs when viruses, bacteria, or other

microbes enter your body and begin to multiply. ... Pathogenic microbes challenge the immune system in many ways.

Viruses make us sick by killing cells or disrupting cell function.

4. **Microbial Identification?**

Microbial identification can be defined as "microbial characterization by a limited spectrum

of tests prechosen and appropriate to the problem being studied"

Methods: Classical or Conventional Method, Serological testing, Nucleic acid based testing.

5. **Write two method? Viral Replication and**

Viral replication involves six steps:

1) ***attachment,***

2) ***penetration,***

3) ***uncoating,***

4) ***replication,***

5) ***assembly,*** 6) ***release.***

... During uncoating, replication, and assembly, the viral DNA or RNA incorporates itself into the host cell's genetic material and induces it to replicate the viral genome.

mmune and Its Types? Define Imunity and

)

Immunity is the balanced state of multicellular

organisms having adequate biological defenses to fight infection, disease,

or other unwanted biological invasion, while having adequate tolerance to avoid allergy, and autoimmune diseases.

Innate Immunity

Plants and animals have what is called innate immunity. Innate immunity is the first line of defense against pathogens.

It involves several cell types, proteins, and even an organ. The organ involved is your skin. Yes, skin is part of the first

line of defense. It protects you and prevents pathogens from getting inside your body.

Adaptive immunity is an important part of the immune system. It is protection from an infectious disease agent that is

mediated by B- and T- lymphocytes following exposure to specific antigen, and characterized by immunological memory.

7. Economies use of fungi?

They play an important role in medicine yielding antibiotics, in agriculture by maintaining the fertility of the soil and

causing crop and fruit diseases, forming basis of many industries and as important means of food.

8. Gram Staining Procedure?

Crystal violet is added. – Primary stain • Crystal violet is washed off. • Iodine is added to enhance binding. – Mordant.

• The slide is washed off with alcohol. – Decolorizing agent • Gram positive bacteria retain crystal violet • Gram negative bacteria appear colorless. • Crystal violet and iodine make a complex in the cytoplasm • Peptidoglycan layer is thicker

in Gram positive bacteria and CV-I is retained. • Alcohol is rinsed off and stained with safranin. – Counterstain • The

smear is washed again. • Blotted dry and examined microscopically.

9. Difference Simple and Differential Media?

2

Selective and differential media are used to isolate or identify particular organisms. Selective media allow certain types

of organisms to grow, and inhibit the growth of other organisms. ... Differential media are used to differentiate closely related organisms or groups of organisms.

10. Horizontal and Vertical Gene Transfer? 3

Vertical gene transfer – Parents to offsprings

Horizontal – From cells to cells within the same generation – From bacteria to bacteria.

11. Difference LB Broth and LB Agar?

LB Broth: broth (LB) is a nutritionally rich medium primarily used for the growth of bacteria. Its creator,

Giuseppe

Bertani, intended LB to stand for lysogeny broth, but LB has also come to be commonly referred to as Luria broth,

Lennox broth, orLuria-Bertani medium.

LB Agar:

Luria broth (LB) is a nutrient-rich media commonly used to culture bacteria in the lab. The addition of agar to LB

results in the formation of a gel that bacteria can grow on, as they are unable to digest the agar but can gather nutrition from the LBwithin.

12. Chemical used in Gram staining?

Both gram-positive and gram-negative cells have peptidoglycan in their cell walls, so initially, all bacteria stain violet.

Gram's iodine (iodine and potassium iodide) is applied as a mordant or fixative. Gram-positive cells form a crystal

violet-iodine complex. Alcohol or acetone is used to decolorize the cells.

13. Viruses, Virion, and Virusoid and Prion?

Viruses: Acellular organisms – Proteins and nucleic acids

Viroids: composed only of circular ssRNA (potato spindle tuber viroid)

Virusoids: Contain circular ssRNA, need helper viruses for replication and encapsidation; also called satellite viruses Prions: Infectious proteins

14.**Uses of UV in Lab?**

- Transilluminators. Used for the visualization of molecular samples, transilluminators are indispensable tools for researchers at life-sciences laboratories. ...
- ***UV Crosslinkers. ...***
- ***Fluorescence Analysis Viewing Cabinets and Work Stations. ...***
- ***Ultraviolet Lamps. ...*** • ***Digital Radiometers/Photometers***

15.Epidemiology and Its type?

Monitoring diseases in a population • Detection of outbreaks and developing epidemics • Implementation of appropriate control measures • Surveillance studies.

There are four most common types of epidemiological studies:

- **Cohort Study.**
- **Case Control Study.**
- **Occupational Epidemiological Study.**
- **Cross-Sectional Study.**
- Epidemiologists use statistical methods to determine whether the differences they see are real or due to chance fluctuations.

16.Streaking Plates method?

Streak plate technique is used for the isolation into pure culture of the organisms (mostly bacteria), from mixed population. The inoculum is streaked over the agar surface in such a way that it "thins out" the bacteria. Some individual

bacterial cells are separated and well spaced from each other.

17.Process of bacterial identification?

The identification of bacteria is a careful and systematic process that uses many different techniques to narrow down

the types of bacteria that are present in an unknown bacterial culture. It produces benefits for many aspects of the

research of microorganisms and helps physicians correctly treat patients.

18.Physical Method of Sterlization?

Heat Method of Sterilization

This is the most common method of sterilization. The heat used kills the microbes in the substance. ... In heat sterilization process, the longer the exposure to heat the better is the sterilization at a given temperature.

19.Disadvantage of Chlorine Sterilization?

disadvantages as well; First of all high concentration of this agent corrodes metals and also damages cloths. Secondly

the strength of this solution decreases with time so whenever you want to use it always prepare its fresh solution.

20.Advantage of dry heat Sterilization?

Advantages of dry heat sterilization include: it is inexpensive, it does not pollute or cause toxic fumes, and it does not

corrode or rust metal objects.Disadvantages include: it is relatively slow and many objects cannot withstand the high temperatures.

21.Virulence

The extent of pathogenicity is called virulence of a pathogen.

22.Name the three subgroups of spiral bacteria

These are curved shaped bacteria. They are further divided into 3 more subgroups.

a. ***Vibrio: curved rods***

b. ***Spirillum: Helical but rigid***

c. ***Spirochete: Helical but flexible***

Spirochetes move by axial filaments which are enclosed by the outer membrane

23.flagellum and function (2

Flagellum: Long filamentous structure that propels bacteria It helps the bacteria in movement.

24.Name the Types of Bacteria on the basis of Oxygen Requirement

a. **Obligate Aerobes**: Do not grow without oxygen. It is necessary.

b. **Facultative Anaerobes:** Oxygen is necessary but start become anaerobes in its absence and can grow without of O2.

c. **Obligate Anaerobes: C**annot grow in the presence of oxygen.

d. **Aerotolerant aerobes:** Do not need O2, also do not bother its presence

e. **Microaerophiles**: Aerobes but need small amount of oxygen. Cannot grow if available in large amount.

25.conjugation, transformation and transduction (3) Transformation

Naked DNA in solution is transferred from one bacterium to another during this process.

Conjugation

Genes transfer process can be mediated by plasmids. Plasmids are extra chromosomal circular DNA fragments that

replicate independently of bacterial chromosome. Conjugation requires direct cell to cell contact and cells have to be opposite mating types.

Transduction

Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a

bacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is

called a capsid) in which viral DNA is packed before the virus is released.

26.how disease can be transmitted (5)

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils,

pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry organisms that

may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets.

Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include

blood and other body fluids, drugs, and intravenous fluids.

What are Vectors? A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another.

Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get

transmitted passively from one location to another. Biological Transmission: Microbes multiply in the insect and is

transmitted actively by insect bites from one individual to another

Any two microbial identification method (5)

Classical or Conventional Method

Serological testing

Nucleic acid based testing … plz See detail on 49 to 54 lessons on handouts **epidemiology and methods (5)**

The study of where and when diseases occur and how they are transmitted in a population

Descriptive Epidemiology: (recording the data about disease). It includes location and time of cases of disease,

gender, health, age of patients etc. (Basically, frequency and distribution of risk factors in population are recorded).

Prevalence and incidence of disease are typical examples. Such studies can be retrospective (after they occur) or

prospective (before they occur). So, in summary, in descriptive epidemiology, information about an already existing disease is recorded.

Analytical Epidemiology: It relates to determining the cause of the disease. Analysis of data, mode of transmission of

diseases and means to prevent diseases also come in this discipline. Such studies can be done with case control

methods in which factors that have preceded the disease are determined. A group of people who have the disease is

compared with another group of people in the same location who are free of the disease. A Cohort method tracks two

groups forward from exposure to outcome. Cohort study compares the experience of a group exposed to the factor

with another group not exposed to the factor. An association between exposure and its outcome is determined which

makes it easy to see the risk factor involved. In other words, cohort studies begin with a group of people (a cohort)

free of disease. The people in the cohort are grouped by whether or not they are exposed to a potential cause of

disease. The whole cohort is followed over time to see if the development of new cases of the disease (or other outcome) differs between the groups with and without exposure.

Experimental Epidemiology: A hypothesis relating to a disease is tested in this approach. For example, smoking cause

cancer can be tested by designing an experiment in mice in which they are exposed to certain levels of smoke over

time and then the outcome is observed. Another example will be testing of a drug for prevention of a disease.

29.Magnetostome (5)

Inclusions of iron oxide

Surrounded by invaginations of plasma membrane Present in G negative bacteria

Act like a magnet. Bacteria can stick to iron containing rocks for nutrition.

Decompose H2O2 which is toxic for cells.

30.Define immunity, difference between innate and adaptive immunity in Detail (10)

Immunity: Ability of the body to protect itself from harmful effects of diseases whereas Susceptibility is the lack of

immunity which provides the pathogens easy establishment to cause the diseases Types of Immunity Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system

against these entire pathogens acts in a similar fashion.
In other words, the same macrophage can phagocytize E. coli,

Staph or Salmonella at the same time (if these pathogens are present there).

Does Not Have Memory. Do not increase in responses to the repeated exposure to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the

entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytes and their secretions provide a second line of barriers for microbes. So, if microbes

succeed in breaking the first barrier, macrophages, neutro-phils and such other cells phagocytize these microbes and

kill them before these can establish infection in the body.

Adaptive (Acquired): It is the third line of defense. It is acquired by experience and specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the

body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened

immune response against E. coli because it has memory cells developed in the body which upon second stimulation

proliferate quickly and mount a greater defensive response.

Increases with repeated exposure: Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes) and Cellular immune response (T cells are made which kills microbial harboring cells)

31.cell membrane function

a. Selective permeable barrier

b. Passive and Active Transport

c. Respiration in microbes (bacteria)

d. Photosynthesis in microbes

e. **Lipid synthesis**

f. Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.

g. PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

Glycoclyx, its composition and function

It is the outermost layer of the cell wall, viscous and gelatinous in nature that surrounds the cells. It is composed of

polysaccharide and polypeptide or both. If glycocalyx is organized and firmly attached to the cell, it is called a

capsule. – Plays important role in virulence (degree of pathogenicity) – Can also be a part of the vaccine against the bacteria to which it belongs.

Glycocalyx is also part of biofilms that bacteria make to attach to surfaces. Bacteria secrete extracellular polymeric substance (EPS) Biofilm protects cells within it. It also facilitates communication amongst cells

If glycocalyx is loosely attached to the cell, it is called slime.

Differentiate between vertical and horizontal gene transfer and name three horizontal gene transfers

Vertical Gene Transfer:

This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has.

Horizontal Gene Transfer:

When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer.

This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer

involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or

recombinant DNA. The cell that has this recombinant DNA in it is called

a recombinant cell. ***Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:***

1. ***Transformation***

2. ***Conjugation***

3. ***Transduction***

34.three types of passive movement

Diffusion Facilitated Difussion Osmosis

35. Economic benefits of fungi

Saccharomyces are used for alcoholic beverages

under anaerobic conditions and if incubated aerobically, it is used in

bread making because it metabolizes glucose to produce CO2 (some ethanol is also produced but gets evaporated

during baking) which expands the dough causing it to rise.

This yeast is also extensively used for molecular biology work and vaccine production.

Vitamin C is obtained from Aspergillus niger.

Cellulase from Trichoderma

Antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products.

Entomophaga (kills gypsy moth), hence is used as a biological pest control.

Mushrooms can be cooked and eaten as a source of proteins for humans.

36.Pili and its functions

These are hair-like structure composed of pilin,

usually one to ten in number. Longer than fimbriae

Used for attachment to: Host cells

Bacteria

Used for DNA transfer from one bacterium to another during Conjugation (Sex pili)

Also function in twitching Motility

Gliding Motility is also the function of the pili.

37.physical nature of media 2

- ***Solid Media***

- ***Liquid Media***

38.germination of spore, 5

Exosporium: A thin delicate outermost covering of the spore

Spore coat: 2 nd layer underneath the exosporium. It is thick and composed of several protein layers. Resistant to chemicals

It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes

favorable for their growth. Cortex: It is the 3rd layer from outside in. It has peptidoglycan in it.

Spore cell wall or core wall: Surrounds the protoplast or spore core Spore core: Contains nucleoid and ribosomes

39.Give two advantages of solid media in bactrial culturing.2

Solidification of the medium is done for purifying organisms from each other as solid medium provides surface for

individual colonies to grow well separated from each other.

Secondly, the solid media are used to study the colony characteristics.

40.Types of microscopy 5

- ***Dark field microscopy***
- ***Phase contrast microscopy***
- ***Fluorescent microscopy***
- ***Transmission electron microscopy***
- ***Scanning electron microsc***ope

41.Writes stages of respiration of glucose 3

Glycolysis: Oxidation of glucose to pyruvic acid

Krebs cycle: Oxidation of acetyl CoA to CO2

Electron Transport Chain: Coenzymes that carry electrons from Krebs cycle or glycolysis are oxidized to create ATP.

42. Differentiate between horizontal and vertical gene transfer.

• Vertical Gene Transfer o This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has. • Horizontal Gene Transfer genes are transferred from cells to cells within the same species, the process is called horizontal gene transfe involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this

genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a

recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:

1. Transformation 2. Conjugation 3. Transduction

43. Define buffers.

o Buffers are used to maintain desired pH.

o Bacteria grown in lab produce acids. o Growth inhibited by acids. o Peptones and amino acids are used as buffers. o

Phosphates are also used for buffers.

44. Define immunity, innate immunity and adaptive immunity.

- **Immunity**: Ability of the body to protect itself from harmful effects of diseases.

- **Innate (Natural):**

o Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the

reason, it is non-specific. o Rapid to respond: Because it is already there, it interacts with the microbes as soon as they

enter the body. o Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body

defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can

phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there). o Does not increase in

response with repeated exposures to the same agent. Therefore, it has no memory.

o First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent

the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside. o Second line

of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in

breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill

them before these can establish infection in the body. Adaptive Immunity:

The adaptive immune system, also known as the acquired immune system or, more rarely, as the specific immune

system, is a subsystem of the overall immunesystem that is composed of highly specialized, systemic cells and processes that eliminate pathogens or prevent their growth.

Adaptive (Acquired): It is the third line of defense.

Acquired by experience, specific to an organism.

45. What is epidemiology? Describe its tyes.

o The study of where and when diseases occur and how they are transmitted in a population. There are three basic types of investigations relating to epidemiology:

Descriptive Epidemiology

Analytical Epidemiology

Experimental Epidemiology:

46. Enlist microbial identification methods.

Classical or Conventional Method

Serological Methods Nucleic

Acid based Methods

Write economical uses of fungi.

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated

aerobically (for bread making), they metabolize glucose to produce CO2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

47. What is sporulation?

• Formation of spores takes place within a vegetative cell and the process is called sporulation or sporogenesis. It is initiated when nutrients become unavailable.

49. Differentiate between viable and direct count.

50. What is flagellum? Write its functions.

Flagellum: Long filamentous structure that propels bacteria

51. What is meant by gene transfer? Name the types.

Vertical Gene Transfer

Horizontal Gene Transfer

1. **Transformation 2. Conjugation 3. Transduction**

52. Describe classification system proposed by Robert H Whittaker.

In 1969, five kingdom classification was proposed by Robert Whittaker as under:

•Plantae: plants

•Animalia: Animals

•**Fungi: Yeasts, molds and mushrooms**

•**Protista**: These are unicellular eukaryotes. Organisms that do not fit into any other category are placed in Protista. They

are larger than prokaryotes. They include algae, protozoa, slime molds and water molds.

•Prokaryotae or Monera: Bacteria

53. What is taxonomic hierarchy?

All organisms can be grouped into a series of

subdivisions that make up the taxonomic hierarchy.

Specie:

A bacterial species represents —a monophyletic and genomically coherent cluster of individual organisms that show a

high degree of overall similarity with respect to many independent characteristics, and is diagnosable by a discriminative phenotypic property‖ Genus

Order

Family

Group

Class

Phylum

Kingdom

Domain

54. What is western blotting?

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial proteins can be

separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzymetagged

antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) is present on the gel.

Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies

55. Enlist types of passive transport.

Simple Diffusion

Facilitated Diffusion

Osmosis

56. Name types of spiral bacteria.

These are curved shaped bacteria. They are further divided into 3 more subgroups. Vibrio: curved rods

Spirillum: Helical but rigid Spirochete:

Helical but flexible

57. Define passive movement.

Passive Movement

This means that substances will move from

higher concentration of a substance to lower concentration. Gradient provides the force for movements of molecules and this happens with no energy expenditures.

58. What is function of lysozymes?

A lysosome is a membrane-bound organelle found in nearly all animal cells. They are spherical vesicles that contain

hydrolytic enzymes that can break down many kinds of biomolecules. Lysosomes are organelles that contain digestive

enzymes. They digest excess or worn out organelles,

food particles, and engulfed viruses or bacteria.

What are fimbriae? Give location and function.

Hair-like small appendages on G- around the entire

cell epithelial cells before they can secrete biofilm.

• Fimbriae also help attach to epithelial cells e.g Neisseria gonorrhoeae

60. Name types of culture medium on the basis of function.

a. **Differential medium**

b. **Selective medium**

c. **Enriched medium**

d. **Basal medium**

e. **Reducing medium**

f. **Selective medium**

g. **General purpose medium**

61. **Write functions of cell wall, flagella and ribosomes**

Cell wall gives support and protection against plasmolysis.

Flagella help bacteria in movement.

Ribosomes are organelles that assist in protein synthesis during translation. **What are types of microscopes?**

a. **Light microscope**

b. **Electron microscope**

c. **Scanning electron microscope**

d. **Transmission electron microscope**

e. **Dark field microscope**

f. **Bright field microscope**

g. **Fluorescent microscope**

63. What do you know about types of techoic acid?

G-positive cell wall contains teichoic acid which may be divided into wall teichoic (remains within the cell wall) acid and lipoteichoic acid (is inserted into the cell membrane).

64. **Write characteristics of integral proteins.**

. Integral proteins act as channels or carriers in facilitated diffusion. Integral proteins are called transporters or permeases ii. No energy is required during transportation of substances through these integral proteins. iii. Two kinds of transporters are known: i. Nonspecific transporters
1. Ions ii. specific transporters **Change in shape 2. Sugars, vitamins**

65. **What are benefits of solid medium?**

Solidification of the medium is done for purifying organisms from each other as solid medium provides surface for

individual colonies to grow well separated from each other. A well isolated colony is assumed to be the progeny of a

single cell. Secondly, the solid media are used to study the colony characteristics.

66. **Define refractive index.**

Refractive Index: Ability of a medium to bend the light o Light

rays move in a straight line through a single medium o Light rays bend when the pass from one medium to another

67. **Define medical microbiology.**

Medical microbiology is a branch of medical science concerned with the prevention, diagnosis and treatment of

infectious diseases. In addition, this field of science studies various clinical applications of microbes for the improvement

of health.

70. **Define virulence. Virulence:**

The extent of pathogenicity

71. **Write methods for 1. Plate Counts**

a. **Pour Plate Method:**

b. **Spread Plate Method:**

2. ***Filtration:***

3. ***The most Probable Number Method:*** 4.

Direct Microscopic Count

72.Write steps of germination of spores.

• Spore germinates when it finds a conducive environment for its growth. **Germination has 3** phases or stages:

Activation: It prepares the spore for germination.

onment.

Germination: Spore starts swelling and loosing its coats etc.

Outgrowth: New components are made.

73. Write types of lipopolysaccharides. 2 Define culture, media and inoculum.

Microbial growth in the lab is called a culture.

A nutrient material that supports the growth of microbes in the lab is called a culture medium.

Microbes introduced into a culture medium that initiate growth of organisms are known as inoculum.

75. Define antiseptics.

Compounds that are used to remove pathogens from living tissue are known as antiseptics.

76. Define phosphorylation.

The addition of an inorganic phosphate group to a chemical compound is called phosphorylation.
e.g conversion of ADP into ATP in living tissues.

Explain phases of growth curve.

In a batch culture, organisms go through various phases of growth. If we follow this growth with respect to time, and

make a graph of this growth pattern, it will give us what we call as a growth curve. There are four distinct phases of this

c**urve. 1-Lag phase 2- Log phase 3- stationary phase 4- Death phase.**

77. What is high frequency of combination?

In some cells that carry F+, F+ (Fertility factor) gets incorporated in the chromosomal DNA which converts F+ cells
to high

frequency of recombination cells (Hfr cells). When conjugation occurs between an Hfr cell and an F- cell, the Hfr cell's

chromosome (with its integrated F factor) replicates, and a parental strand of the chromosome
is transferred to the

recipient cell. Since this transfer starts from the middle of F factor gene, and most of the time, this transfer is not

complete; the whole chromosome of Hfr cell is not transferred. However, some genes can be transferred during this

process. When these genes become integrated into the genomic DNA of the recipient cells, the recipient cells acquires

new versions of genes that were not part of its genome previously. It may be noted that F- cells remains F- negative cell

because F factor is not transferred completely. This whole process is illustrated in the accompanying diagram.These newly

acquired genes can be mapped easily with respect to the time they get transferred to F- cells.

79. Differentiate between clone and strain?

Clone: Population of cells derived from a single cell that are genetically identical.

Strain: A genetic variant of a clone is called a strain.

80. Differentiate between positive sense and negative sense RNA.

81. What is zoonosis?

Zoonosis: Primarily an animal infection, but can be transmitted to humans. It can assume epidemic proportions in humans and is difficult to control or eradicate.

82. What are interferons? Give their function.

Interferons are antiviral proteins, produced by macrophages and lymphocytes in response to viral infections. Fibroblasts

can also secrete it. These proteins interfere with viral replication. Thereare two types of interferons:

Species specific: This means that interferon produced by humans cannot be used in dogs or cats or vice versa.

Not viral specific: This means that interferons produced in response to virus A can also be effective against virus B.

83. **Explain characteristics of aerophiles.**

84. ***Describe conjugation, transduction and transformation. Horizontal gene 10***

85. ***transfer:***

When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This

can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome

becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a

recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:

1. ***Transformation 2. Conjugation 3. Transduction***
1. Transformation o Naked DNA in solution is transferred from

one bacterium to another during this process. Frederick Griffith observed this process in 1928 when he was working with

Strep pneumonia in his lab. This bacterium has two strains: one is capsulated and is called a Smooth strain (which
is

pathogenic) and the other is Rough strain (nonpathogenic).

o When live encapsulated bacteria are injected into a mouse, the mouse gets killed by the microbe. When rough strain is

injected, the mouse is not killed. Moreover, when heatkilled encapsulated bacteria are injected, it also does not kill the

mouse. However, when heat-killed strain is mixed with rough live strain, and a mouse is injected with this mixture, the

mouse dies. If you isolate the bacterium from these dead mice, the bacteria resemble with encapsulated strain of Strep

pneumonia. The interpretation of this experiment led to the mechanism that rough live bacteria take up DNA from the

heat-killed encapsulated strain and becomes pathogenic by acquiring the ability to form a capsule. Very few bacteria have

this property of accepting foreign DNA and most of it is degraded (only small fragments of DNA get incorporated in the

genome of the recipient. The cells that can take up DNA are called competent cells. We can create such microbes in the

lab that can accept a given DNA during gene transfers that we undertake in our DNA work. The cells that take up the

DNA and incorporate this in their genome successfully are called transformed cells.

2. Conjugation o Genes transfer process can be mediated by plasmids. Plasmids are extra-chromosomal circular

DNA fragments that replicate independently of bacterial chromosome. In other words, a given bacterial cell can have

multiple copies of plasmid with the same cell. Plasmids often have genes that are not strictly needed by the microbe for

its growth. However, the genes on the plasmids provide some advantages to the bacterium. For example, antibiotic

resistant genes offer a mechanism for the bacterium to resist the

presence of that antibiotic. In other words, bacterium that has a plasmid with an antibiotic gene on it can still multiply in the presence of that antibiotic.

o Conjugation requires direct cell to cell contact and cells have to be opposite mating types. Plasmids in G – ive

cells have genes that code for sex pili because pili are needed to make a bridge between the two mating type cells. Since

there is no pilus present in G +ve bacteria, they secrete sticky surface molecules

instead which create contact between the two cells. The pilus along with two mating cells is shown in the diagram below:

o ***These plasmids are transmissible from one***

bacterium to another during conjugation.

o ***Plasmid is replicated and a single copy of the plasmid is transferred to the recipient bacterium through the pilus. 3***

o ***A complementary copy is then made in both the cells to make the DNA of the plasmid double stranded molecule.***

o In E. coli, fertility factor (F factor) plasmid was the first
plasmid observed to be transferred from one organism to

the other, hence those bacteria that have this plasmid are called F+ cells to differentiate from those that do not have one (F-

). However, once F- cells acquire F+ plasmid, they also become F+.

3***. Transduction o Transferring of***

a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a

bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which

viral DNA is packed before the virus is released. What happens is that when phages cause lysis of infected cells, bacterial

chromosomal DNA also gets fragmented and such fragments can sometimes be packed into the viral capsids much like the

viral DNA. Now, the capsid carrying the bacterial gene or genes in that fragment can infect another E. coli and can inject

its DNA (bacterial DNA fragment) into it resulting in the transfer of genes from one bacterium to another.

85. Explain methods of microbial control in detail.

These methods can be broadly classified into four groups: Physical, Mechanical, Chemical and Biological

Physical methods of microbial control

Heat: It is the cheapest sources of all and easily available to control microbial growth. There are a few heat related

concepts that we must appreciate. Heat denatures enzymes.

Thermal Death Point: It is the lowest temperature at which all cells in a culture are killed in 10 min.

It will be a specific temperature for a specific species of organism.

Thermal Death Time: Time (minimum) during which all cells in a culture are killed at a given temperature. This will

vary from temperature to temperature for the same organism. Obviously, higher temperatures will take less time to kill the organisms than low temperatures. rther be classified into dry and moist heat methods.

Heat kills by oxidation

Form of dry heat include:

1. ***Dry heat (hot-air oven),***

2. ***Flaming (Platinum loop sterilization in the flame of Bunsen burner),***

3. ***Incineration (burning to ashes)***. oC, and

Tyndallization (100oC for 3

consecutive days to sterilize sugar solution that can be degraded by autoclaving) and pasteurization (usually at low temperature than 100oC).

Pasteurization: This technique typically employs low heat for killing pathogenic and food spoiling bacteria in milk.

However, it does not kill thermoduric organisms. Thermoduric organisms are not pathogenic though. Products other than

milk, such as ice cream, yogurt, and beer, all have their own pasteurization times and temperatures, which often di er considerably. Dairy industry tests phosphatase to determine if pasteurization of dairy product has occurred effectively because phosphatase present in raw milk gets inactivated with effective pasteurization temperatures. The following three equivalent heat treatments can be given to achieve pasteurization of milk: o 63°C for 30 min o o High-temperature shorttime: 72°C for

15 sec o Ultra-high-temperature: 140°C for 4 sec

Other Physical Methods

Low temperature inhibits microbial growth by slowing down chemical reactions.

Refrigeration: Materials stored at 2-8°C can last for a day or two.

Deep-freezing: Long term storage of materials is possible at minus temperatures in the deep freezers.

Lyophilization: Freeze drying is another method for long term storage of food and other materials.

High pressure deshapes and denatures proteins: This technique is used for preserving juices and wines.

Desiccation prevents metabolism: This method is very old and still in use today. Food stored in high concentrations of salt or sugar solution can keep for long.

Osmotic pressure causes plasmolysis. High salt concentrations for example can take the water content of the cells out leaving them starved of water.

Filtration: It is also a physical method. It can be used to filter air (using high efficiency particulate air filter) or liquid medium using membrane filters.

Radiation: o Ionizing and non-ionizing radiation o Ionizing Radiation: High energy waves that destroys microbes.

• Ionization radiations include gamma rays, X rays, or high-energy electron beams. The ionizing radiation

possesses a wavelength shorter than that of nonionizing radiation, less than about 1 nm. All these short wavelengths cause

ionization of water generating high reactive hydroxyl radicals which are damaging to the cells as they bind to DNA and

results in mutations. Medical plastic supplies, medicines and meat products can be sterilized by radiation.

-ionizing Radiation: wavelengths longer than 1nm fall into non-ionizing radiation. The best example is UV light.

UV light causes mutations in the DNA by dimerization of thymidine bases. These thymine dimers inhibit correct

replication of the DNA during reproduction of the cell. The most useful wavelength is 260 nm. This wavelength is

specifically absorbed by DNA. UV radiation is used to control organisms in the air mostly. However, surfaces are also

sterilized by UV radiation. UV light cannot penetrate deep into the fluids, so only surface bacteria are killed.

ETHYLENE OXIDE FOR MICROBIAL CONTROL

o Ethylene Oxide: Many heat-sensitive items such as disposable plastic, petri dishes and syringes, heartlung machine

components, sutures, and catheters are sterilized by ethylene oxide. Ethylene oxide is a strong alkylating agent that kills

by reacting with functional groups of DNA and proteins to block replication and enzymatic activity. It rapidly penetrates

packing materials, even plastic wraps. EtO is explosive, supplied in a 10% to 20% concentration mixed with either CO2 or

dichlorodifluoromethane. A clean object can be sterilized if treated for 5 to 8 hours at 38°C or 3 to 4 hours at 54°C when the relative humidity is maintained at 40 to 50% and the EtO concentration at 700 mg/L. CHEMICAL METHODS FOR MICROBIAL CONTROL

Chemicals are used to control microbial growth. Factors that influence the efficacy of a chemical disinfectant include the

concentration of the disinfectant, presence of organic matter in the environment (matrix) where the disinfectant is being

used, pH of the environment (disinfectants are more effective at acidic pH), and time of exposure to the disinfectant.

86. How infectious diseases are transmitted? Explain in detail.

1) **how infectious disease can be transmitted explain it briefly transformission occur to contact ,vehicle n vector.(10)**

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils,

pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example.

These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze

may produce 20,000 droplets. Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include

blood and other body fluids, drugs, and intravenous fluids.

Vectors o A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. o

Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.

Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

Nosocomial Infections o Hospital acquired infections are called nosocomial infections.

87. Explain salient features of viruses, virions and viroids.

VIROIDS

These are infectious agents that consist only of RNA and found in plants only. No capsids, no envelopes, just RNA. They

are covalently closed circular ssRNAs about 250 nucleotides long. The circular RNA normally exists as a rodlike shape

due to intrastrand base pairing, which forms doublestranded regions with single-stranded loops. They do not encode any

gene products, so they are replicated by cellular DNA dependent RNA polymerase. A plant infected with such viroid

may not show any signs of illness. However, in other plants, they may cause severe disease. They cause the disease by

RNA silencing process.

VIRUSOIDS o They are similar to viroids; however, they encode one or two gene products and need a helper virus to

infect the host cells. The helper virus supplies gene products and other materials needed by the virusoid for completion of its replication cycle. Hepatitis D

virusoid is a typical example which uses Hepatitis B virus as a helper. PRIONS

o These are proteinaceous infectious particles that cause neurodegenerative diseases such as scrapie in sheep, bovine

spongiform encephlopthy, and Kuru in humans. These are abnormal form of cellular proteins. How these proteins

accumulate in the cell is not clear, although, some genetic components are known to be involved.

88. Explain methods of microbial identifications.

89. How fungi can be identified by biochemical testing?

Yeast if grown on solid medium grows like bacteria. In other words, colonies of yeast look like bacterial colonies but are bigger in size.

carbohydrates aerobically and produces CO2 and water; however, if oxygen is denied, carbohydrates get converted fermentatively into ethanol and CO2.

90. IDENTIFICATION OF SELECTED FUNGI i are

cultured using a

slide (slide culture) on top of Sabouraud's agar. Spores are placed on top of the medium or embedded in the medium. A block of the medium can be used to grow fungi or a

glass coverslip can be used to grow spores into hyphae and mycelium

and spore arrangements can be studied under the microscope by staining with lactophenol cotton blue. Slides stained with

lactophenol cotton blue can also be seen under the microscope.Stained slides can show specific arrangements of spores,

their sizes and other morphological features as seen in the figure for Aspergillus

92. ***Write functions of cell membrane of bacteria.***

- ***Selective permeable barrier***
- ***Passive and Active Transport***
- ***Respiration in microbes (bacteria)***
- ***Photosynthesis in microbes***
- ***Lipid synthesis***
- Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.
- PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

93. Write names of 5 types of bacteria with respect to temperature.

Organisms are basically classified into three groups based on the temp requirements:

1. ***Psychrophiles***: These are further divided into strict psychrophiles and psychrotrophs: o Psychrotrophs: Cold loving: 15 0C o ***Psychrotrophs***: Optimum temp is 20-30 0C. Food spoilage bacteria that can spoil food during refrigeration.

2. ***Mesophiles: 25 – 40 0C***: These are the ones that cause diseases in animals and humans.

o Moderate temp loving organisms o Optimum: 37 0C as this is the body temperature of humans and animals.

3. ***Thermophiles: 50 – 60 0C***

o Heat loving o Important in organic compost piles o These are further divided into hyperthermophiles that grow

optimally at 80 0C. They live in hot springs. Every group of these organisms has a:

o Minimum growth temp o Optimum growth temp. o Maximum growth temp.

94. Write characteristics of metachromatic granules.

Metachromatic Granules

- Also called volutin, they stain red with certain dyes such as methylene blue. That is why they are called metachromatic

(stain in different color as methylene blue gives blue color but the color on these granules is red).

• Large inclusions •

These inclusions contain inorganic phosphates. Inorganic phosphates are used up in ATP synthesis.

•

Characteristics of

Corynebacterium diphtheria: This bacterium can be identified by the presence of these granules in it.

95. Explain five kingdom classification systems.

In 1969, five kingdom classification was proposed by Robert Whittaker as under:

•***Plantae: plants***

•***Animalia: Animals***

•***Fungi: Yeasts, molds and mushrooms***

•***Protista: These are unicellular eukaryotes.*** Organisms that do not fit into any other category are placed in

Protista. They are larger than prokaryotes. They include algae, protozoa, slime molds and water molds.

Prokaryotae or Monera: Bacteria and Archea

96.Explain layers of endospore.

Endospore and its various parts/structures

- ***Exosporium***: A thin delicate outermost covering of the spore
- ***Spore coat***: 2nd layer underneath the exosporium. It thick and composed of several protein layers.

Resistant to chemicals

It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes favorable for their growth.

- ***Cortex***: It is the 3rd layer from outside in. It has peptidoglycan in it.
- ***Spore*** cell wall or core wall: Surrounds the protoplast or spore core
- ***Spore core: C***ontains nucleoid and ribosomes

97. Enlist types of bacteria on the basis of oxygen.

Based on the use and concentration of Oxygen, microbes can be classified into five groups:

Obligate Aerobes: Oxygen must be present for their growth. These organisms do not grow if O2 is not present in the

medium. If you test these organisms in a tube containing nutrient agar, the growth will remain confined to the surface only

(see the diagram for the concept). *Mycobacterium tuberculosis is an example of such an organism.*

Facultative Anaerobes: These organisms prefer aerobic respiration; however, if oxygen is not available, they can use anaerobic respiration or fermentative modes for generating ATPs. These organisms will be present throughout the tube,

however, more growth will be seen on the top (close to the surface) because these organisms prefer using oxygen if it is available.

Obligate Anaerobes: Although, these organisms do not use oxygen, they cannot grow in the presence of oxygen. The

reason is that oxygen creates toxic compounds which can kill these cells. Normally, oxygen toxic compounds are

eliminated by the cells, however, strict anaerobe do not have a system to get rid of these toxic compounds when they are

made in the cells. So using or not using oxygen is one thing and growing in the presence of oxygen (and not utilizing it) is

another. The growth will remain confined to the bottom which has little oxygen.

o ***Aerotolerant Aerobes***: These organisms do not use oxygen and also are not bothered by the presence of oxygen. In

other words, they have a better system to dispose of toxic oxygen compounds. They will grow evenly throughout the tube of medium

98.Name five types of microscopy . 5 marks ***Dark field microscopy***

o This microscope makes use of a special condenser that provides illumination from the sides only as depicted in the diagram below:

o This special arrangement of the condenser throws light rays at an oblique angle so that if the specimen is not present on

the slide, there will be no image formed. However, if the the organisms are present on the glass slide, those organisms

will reflect the light into the objective lens and we will see the cells as bright objects in a dark background much like we see stars in the sky at night.

o The beauty of this microscope is that one does not have to stain the organisms to see them under the microscope.

Phase contrast microscopy

This instrument is built on the concept of phases of light waves. The light ray that passes through an organism gets

retarded about ¼ of its wavelengths than the light that passes outside the organism (through the glass slide only). To

further retard the retarded light ray, a phase plate is inserted into the body of the microscope (above the objective lens)

which is thicker in the center than the ends. This arrangement causes the retarded light to further slow down about ¼ of its

wavelength, which when recombines with the unretarded light creates a contrast which makes the cells/organisms visible.

o This method for visualizing organisms also does not involve staining.

Fluorescent microscopy o This microscope is equipped with UV light that provides the source of light for

illumination. However, this light source does not pass through the slide, it instead falls over the slide (having specimen) and creates fluorescence.

o ***Fluorochromes*** are substances that absorb short wavelength of light and emit longer wavelength

(visible light spectrum) o ***Fluorochromes*** are

attached to antibodies that are specific to

organisms. So, if an organism is present in the

specimen,

we can use specific antibodies to determine the presence of those organisms by use of fluorochrome tagged antibodies.

See the accompanying ***diagrams for details. Look for UV lamp (Mercury Lamp). Transmission electron microscope***

o ***Electron microscope (EM) makes use of electron beam***

as an alternative to light rays. Since electron beam has much

shorter wavelength, resolution of EM is 0.1nm compared with light microscope which is 200nm.

o Second difference of EM with light microscope is that focusing of electron beam is done not by glass lenses but by electromagnets that acts like lenses.

o The third difference in EM is that the image of the specimen is formed on a fluorescent screen like a TV screen.

Scanning electron microscope

This is a modified version of electron microscope. Electron beam is allowed to fall on to the specimen and the image is

formed from the reflected electrons that are collected and assembled into an image by the detector. It gives a three dimensional picture.

99. Explain ELISA.

There are some interactions (Antigens and Antibodies) that are not directly evident. Interactions may have taken place,

but there is no visible direct clue if the interaction has taken place or not. For such interactions, we use indirect methods.

Say, there is big body of smoke visible to you from a kilometer ahead of you. What is your guess? Yes, you guessed it

right. There is a big fire there. Did you see the fire? Probably not. You guessed on the basis of smoke that there is a fire

there. This is called indirect way of interpretations. A beautiful example of such a test is ELISA where we do not know if

the interaction has taken place or not. However, we can see it indirectly. We will see that in the following paragraph which discusses ELISA.

Enzyme-Linked Immunosorbent Assay: This test is done by coating the ELISA plastic plates either with the antigen

or the antibodies. Suppose we have Brucella antigen coated ELISA plates that can detect antibodies from the serum of a

Brucella infected animal. You take the serum and put into the wells of ELISA plate. You allow time for antigen and

antibody to interact and then wash these wells with some buffer to remove unbound antibodies. These bound antibodies

are not visible at this step. So, how do we see this interaction? Well, we use another antibody that has been made against

the antibody bound to the antigen. This special antibody has been tagged with an enzyme. After allowing time for

interaction of this 2nd antibody with the 1st antibody, 2nd antibody is also given a washing to remove unbound antibody

molecules. Again, we cannot say if interaction has taken place or not. Here comes an indirect way of detecting this

interaction. Remember that the 2nd antibody has a molecule of an enzyme attached to it. Now, if we provide substrate to the enzyme in the wells, enzyme will convert the substrate into a color compound visible to the eyes. The color change can be measured by a spectrophotometer as well which gives a quantitative data 100. Explain types of conidiophores. 101. Explain types of algae.

- ***Phaeophyta (Kelp):*** Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell

walls, are multicellular and contain chlorophyll a and c,
xanthophylls. These store carbohydrates and are used for algin

which is a thickener in ice creams. Can grow upto 20 cm in one day.

- ***Rhodophyta***: Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and

contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan

(both are used as thickeners of foods).

- ***Chlorophyta***: Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These

algae are believed to have given rise to terrestrial plants.

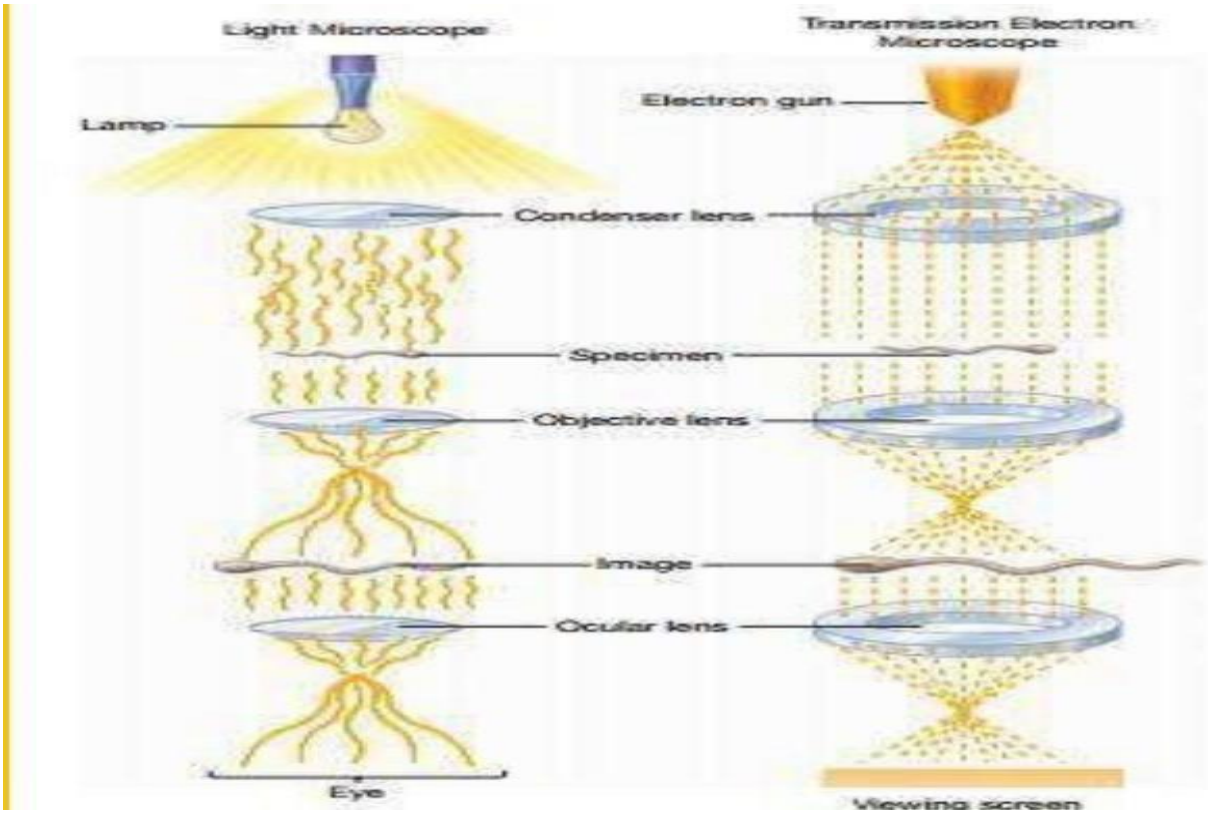
- **Bacillariophyta**: Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and

carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels.
Diatoms: Come in beautiful colors and shapes.

- ***Dinoflagellata***: These are unicellular algae collectively called as plankton (free floating organisms). These cells have

cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll a and c, and

carotene. These algae store starch in them.



Dinoflagellates in the genus Alexandrium produce neurotoxins (called

saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and

clams. Blooms of Alexndrium gives ocean a deep red

color commonly known as red tide. o

Oomycota (Water molds): They are

basically decomposers of dead matter. Superficially,

these resemble with fungi;

however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores are called zoospores.

o Terrestrial members of water molds are also plant parasite. Phytophthora infestans was responsible for Irish potato blight during mid1800s.

P. ramorum causes sudden oak death

Parts of Electron Microscope

1. Which of the following enzyme is used by sense stand RNA virus in its replication_____**RNA Dependent RNA** polymerase
2. The carbohydrates and Protein complex Projecting from the surface of envelopes are called_____**Capsomeres**
3. Neurodegenerative disease are caused by _____**Prions**
4. The infections which are acquired by hospitals are called_____**Latent infections**
5. The first step in virus replication is _____**Attachement**
6. Are normal biota that can cause disease under stress conditions_____**Opportunistic pathogens**
7. The microbial communities that live in or in humans is called_____**micriniome**
8. The study of occurrence of disease and how they are transmitted_____**Epidemiology**
9. Is the number of new cases and existing cases In a given time period in a population Endemic._____ **incidence**
10. Is primarily an animal infection but can be transmitted to human prognosis_____Zoonosis
11. Consistently present but with low incidence of disease is_____**Endemic**
12. Following cell culture can be used for the cultivation of viruses_____**All**
13. The disease which is present worldwide _____**Pandemic**
14. The product of nucleic acid and Proteins occur in which step of replication of virus_____**Biosynthesis**
15. The number of cases in a given time period in a population is called_____ **incidence**
16. The number of peoples effected in relation to the total population in a given time period____ **Morbidity Rate**
17. The disease with incident rate _____**Epidemic**
18. The extent of pathogenicity of an organism is _____**Virulence**
19. The relationship in which both ***organisms benefited*****_____Mutualism**
20. The death from a diseases is called **_____Mortality**
21. **The dose** which can kill 50% of the cells or testing population_____**TCID50**
22. The prevalence of a diseases worldwide is called_____**Pandemic**
23. The incidence of a disease is called_____**Morbidity**
24. The sites where organisms remain Viable and Act as a source of infection_____ **Reservoir**
25. Is a competitive exclusion by competition of nutrients or by Chemicals_____**Microbial Antagonism**
26. **An** organiam with pathogen infected indivisual who does not show any sign or symptoms of disease_____**Carrier**
27. Capsin is made up of protein subunits called _____**Capsomeres**
28. Which of the following are used for culturing virus_____**All of them**
29. The ability to cause disease is called_____ **Pathogenecity**
30. The method of Epidemiology in which hypothesis of a disease is tested _____ ___**Experimental Epidemiology**
31. The relationship in which ***one organism is harmed*****_____Parasitism**
32. In the replication of DNA virus , a portion of viral DNA is transcribed and produces _____**mRNA**
33. Microbes which are permanent resident of human body but do not cause disease_____**Normal microbiota**
34. The time between infection and disease_____**incubation period**

1. Hemoglobin is called oxyhemoglobin when ____ is bound…….. **Oxygen** 2.

The average total lung capacity of holding of air in an adult human male is

about_____..... **6000 mL**

3. The average total lung capacity of holding of air in an adult human female is about_____..... **4200 mL**

4. Alveoli are tiny air sacs in the_____.....**Lungs**

5. Most diving animals have _____ levels..... **High hemoglobin and myoglobin** 6. _____ are located in the medulla of mammals and many other air breathing vertebrates……**. Central Chemoreceptors**

7. Osmolytes are substances which reduce the need to maintain osmotic pressure with _____ ions…… **Inorganic**

8. Pulmonary surfactant is a surface-active lipoprotein complex formed by _____ alveolar cells…… **Type II**

9. _____ is the active regulation of the osmotic pressure of an organism's body fluids…… **Osmoregulation**

10.Lung volumes and lung capacities refer to the volume of air in the lungs at different phases of the_____..... **Respiratory cycle**

11.Thromboxane A is synthesized by _____ and causes vasoconstriction…… **Platelets**

12.Which one of the followings is not a respiratory pigment?..... **Charcoal**

13.A condition in which the body or a region of the body is deprived of adequate oxygen supply is called _____.....**Hypoxic**

14.Blood with hemoglobin can carry 20 ml oxygen per 100 ml at _____ **sea level…… 37°C**

15.Air capillaries are the sites where oxygen and carbon dioxide are _____..... **Exchanged**

16.Do not possess a diaphragm…….. **Reptiles**

17.Peripheral chemoreceptors are located in…… **All of the above**

18._____ oxygen binds with hemoglobin……**97%**

19.Angiotensin II is produced in…… **lungs**

20.Lung ventilation is a mechanical process and it consists of _____ phases…… **Two** 21.A male human can forcefully expel additional ___ air by use of accessory muscles……**1000 mL**

22.A female human can forcefully expel additional ___ air by use of accessory muscles……**700 mL**

23.Lungs are located in…… **Thoracic cavity**

24.Gas transfer system involves _____ basic steps…… **Four**

25.Alveoli are very small……**bubble-like**

26._____ is the phase of ventilation in which air enters into the lungs…… **Inspiration**

27.If all sites on hemoglobin molecule are occupied by Oxygen, blood is _ saturated….**100%**

28.Amount of air that moves into or out of lungs during a single, quiet respiratory cycle is called _____.....**Resting tidal volume**

29.Fishes live in aquatic environment that contains less than _____ of oxygen than present in air……..…**2.5%**

30.Respiratory surface area of human lung is _____..... **50 to 100 m2**

31.Affinity of hemoglobin for CO _____ times greater than for oxygen……..**200**

32.Pulmonary Surfactants have a half-life of about ____ hours in mammals……**12**

33.Respiratory system of birds consists of _____ air sacs……**7 or 9**

34.The bronchioles are an important part of the _____.....**Respiratory system**

35.During Inspiration. the diaphragm……**Contracts**

36.In expiration phase of ventilation air with low _____ and high _____ content is moved out…… **Oxygen, carbon dioxide**

37.Many air-breathing aquatic vertebrates go under water for varying periods of time is termed as…… **Dive**

38.Carbon dioxide transportation in blood occurs in ____ forms…… **Three**

39._____ is a potent vasoconstrictor hormone…….. **Endothelin**

40.Fishes have gills which are efficient respiratory organs to extract ___ in water……**Oxygen**

- Virus is surrounded by a Protein coat Called _____ **Capsid**
- Location of Algae in oceans depends upon _____**All**
- Virusoids are also known as _____**Satellite RNA**
- Viroids only infect _____**Plants**
- Reverse Transcriptase is an Enzyme that transcribes _____**__RNA into DNA**
- Virus Attach to the _____**Host Cell membrane**
- Capsomeres are composed of _____**Proteins**
- When virus is outside a cell in extracellular state is called_____**Virion**
- Plaque Formation is method of Viral _____**Enumeration**
- Taga virus Gets Entry into the host Virus by _____**Pinocytosis**
- Kuru in humans is caused by_____**Prion**
- Route of Entry Effects _____ _____**ID 50**
- Lactoferrin is iron containing protein present in _____**Blood**
- 10 % Tannic acid is used as a_____**Mordant**
- Isoniazid is commonly used to treat _____**Tuberculosis**

- The bursa is an epithelial and lymphoid organ found only in_____**Amphibinans** Thickness of the capsule depends upon the _____**cultural conditions.**
- Blood Transfusion reaction is an example of _____**Type II**
- Quinolones and Rifampin are antibiotics that inhibit_____**DNA Replication**
- Motile and unicellular organisms_____ _____**Protozoa**
- Antibiotic resistant genes can be transferred through _____**Specialized Transduction**
- Spores formed by rounding and Enlargement within a hyphal segment is called_____**Chlamydioconidia**:
- Staining of Fungi is done by using _____**Lactophenol Cotton blue**
- Rabies Virus is an Example of _____**Helical virus**
- The range of ? Of UV is most bactericidal_____**200_280**
- PCR is method of Microbial identification _____**Molecular**
- Study of microbes is called _____**Microbiology**
- Mycobiant contains_____**Fungi**

1. Peripheral proteins_____ loosely connected and soluble in water
2. Plasmid_____ extrachromosomal circular DNA
3. Mesophils optimum temp_____is 37
4. Unicellular fungi is _____yeast
5. Asexual spores are formed by_____ mitosis
6. Mycobiont is_____fungi
7. ELISA is used for_____Antigen and antibody
8. Western blotting is used for_____proteins
9. Consistently present but with low incidence_____endemic
10. Which is not belong to innate immunity_____specific
11. Which is not lymphocyte_____monocytes
12. Which is not belong to ELISA_____Probe
13. Extreme halophiles require high salt concentration_____30%
14. pH of mold is _____ 5-6
15. which do not grow in the presence of o2_____ Obligate anaerobes
16. the process in hot air sterilization is _____ oxidation
17. which is not a method of direct count_____ Spectrophotometer
18. resolution of microscope is_____ 0.2µm

19. persisitent infection _____Viruses replicate slowly

20. cell wall is absent in_____ mycoplasma

21. Resolution of Electron Microscope _____0.2nm

22. Ribosomes made up of _____mRNA and protein

23. Rings of basal body of gram negatives_____ 4

24. Gram negative appears on stain _____red

25. Basic dye methylene _____blue

26. Rat bite fever caused by _____ Spirillum minus

27. High salt concentration used by obligat halophiles_____30%

28. Acellular... viruses

29. Resolution of Electron Microscope....0.2nm

30. Ribosomes made up of... mRNA nd protein

31. Rings of basal body of gram negatives.....4

32. Gram negative appears on stain...red

33. Basic dye.... methylene blue

34. Robrthook draws microbe in.....

35. Rat bite fever caused by.... Spirillum minus

36. High salt concentration used by obligat halophiles.... 30%

37. Small small colonies type name...

38. Direct count method...membrane filter

39. Study of fungi...mycology

40. Standard condition for hot air oven....170*C for 2hours

41. Autoclaving at... ?121*C 15psi 120min

42. EtO do not use...1500mg/ml

43. Optimum pH for molds is...5 -6

44. Peripheral proteins----------loosely connected and soluble in water

45. Plasmid--------extrachromosomal circular DNA

46. Mesophils optimum temp. is _____37

47. Unicellular fungi is yeast

48. Asexual spores are formed by _____mitosis

49. Mycobiont is---------fungi

50. ELISA is used for_____ Antigen and antibody

51. Western blotting is used for _____proteins

52. Consistently present but with low incidence_____ endemic

53. Which is not belong to innate immunity_____ specific

54. Which is not_____ lymphocyte monocytes

55. Which is not belong to ELISA_____ Probe

56. which do not grow in the presence of O2 Obligate anaerobes the process in hot air sterilization is____ oxidation

57. which is not a method of direct count Spectrophotometer persisitent infection --------Viruses replicate slowly 58. cell wall is absent in mycoplasma

59. Hot air oven temperature is ___ 170degree for 2 hour

60. Example of binguanides is _____ chlorohexidine

61. Bacteriophage is virus infecting _____ bacteria

62. Sight of entry through which microorganism enter body____ ? 5.

63. Plasmid is_____ single stranded or double

64. Lime disease is caused by:

65. Proteins are separated by _____ gel electrophoresis

66. Structural support is provided by _____core polysaccharides

67. Cell wall is absent in _____ mycoplasma

68. Alpha hemolysis gives _____greenish halo colour

69. Study of fungi is mycology

70. 30S ribosome contain _____ 2rRNA

71. **70 % alcohol is more effective**

72. **Microbiology definition**

73. Epidemiology

74. Algal body is _____thallus

75. Obligate halophiles live in_____ 30% Salt

76. Specimen or sample is placed on_____ stage

77. Mycobacterium is stained by _____ acid fast staining

78. Pilli transfer DNA from one bacterium to _____other

79. Half of the dry weight of microbe is _____ carbon

80. Hot air method is 121 degree at 15psi for_____ 20min

81. Peripheral proteins are loosly connected and _____ soluble in water

82. Mesophils optimum temperature is ____ 37

83. A sexual spore is formed by _____mitosis or meiosis

84. Mycobiont is fungi

85. Elisa is used for antigen and_____ antibody

86. Consistently present but with low incidence is _____ endemic

87. Which not belong to innate immunity?_____ Specific

88. Which is not lymphocyte? _____ Monocytes

89. Probe not belong to _____elisa

90. Process in hot air sterilization is oxidation

91. Spectrophotometer is not a method of direct count

92. Virus replicates slowly

93. Cell wall absent in mycoplasma

94. Water molecules pass through aquaporines

BT102 FINAL TERM MCQS AND SUBJECTIVE (NOTES)

Created by Aftab Ahmed and Irtaza Ali

1. Hot air oven temperature is _____ **170degree for 2 hour**

2. Example of Biguanides is_____ **chlorohexidine**

3. _____ is virus infecting bacteria. **Bacteriophage**

4. Sight of entry through which microorganism enter body. **Portals of entery**

5. Plasmid is _____ stranded **_____Double____**

6. Lyme disease is caused by. _____**BORRELIA BURGDORFERI**

7. Proteins are separated by_____ **Gel electrophoresis**

8. Structural support is provided by ____.**core polysaccharides**

9. Cell wall is absent in ____.**Mycoplasma**

10. Alpha hemolysis make _____**GREENISH HALO**

11. Study of fungi is_____. **Mycology**

12. 30S ribosome contain _____.**2rRNA**

13. _____ % alcohol is more effective. **70%**

14. Study of microbes is called _____**Microbiology**

15. Monitoring disease in a population. _____**EPIDEMIOLOGY**

16. Algal body is___. **Thallus**

17. Obligate halophiles live in_____. **30% salt**

18. Specimen or sample is placed on____. **Stage**

19. Mycobacterium is stained by ____**acid fast staining**

20. **Pilli_____**transfer DNA from one bacterium to other

21. Half of the dry weight of microbe is **_____carbon___**

22. Hot air method is_____. **121 degree at 15psi for 20min**

23. _____proteins are loosly connected and soluble in water. **Peripheral**

24. Mesophils optimum temperature is_____. **37°C**

25. A sexual spore is formed by___ _____**Mitosis**

28. Consistently present but with low incidence is_____. **Endemic**

29. Which is not lymphocyte____. **Monocytes**

30. Process in hot air sterilization is _____.**Oxidation**

31. Water molecules pass through____. **Aquaporines**

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32. Direct count method _____.**Membrane filter**

33. Extra chromosomal circular DNA ___.**Plasmid**

34. Unicellular fungi is ____.**yeast**

35. Mycobiont is _____.**fungi**

36. ELISA is used for_____. **Antigen and antibody**

37. Western blotting is used for ____.**Proteins**

38. Which is not belonging to innate immunity? ____ **Specific**

39. Which is not belonging to ELISA_____ **Probe**

53. Which do not grow in the presence of O2 ___.**Obligate anaerobes**

54. Which is not a method of direct count_____ **Spectrophotometer**

55. Persistent infection: Virus replicates but _____**slowly**

56. Which of the following is not bacterial gene transfer method_____ **Transfection**

57 Which one is correct answer regarding sterilization_____.**Removel of microbes and their spores.**

58. Ethylene oxide is strongly alkylating agent that kills the microbes reacting with the functional group of _____to block replication and enzymatic activity. ____**DNA**

59. ____ is the transfer of DNA by a virus called bacteriophage.__**Transduction**

60. Transformation is the horizontal gene transfer between two_____. **Bacteria**

61. The primary stain of gram method is_____. **Crystal violet**

62. Antigen antibody reaction is a _____method for microbial identification. _____**Serological**

63. Phenol and phenolic compound is widely used antiseptic for disruption .**Plasma membrane**

64. Buffer are used to maintain ____in medium. ____**pH**___

65. Protains could be detected by ._____ **Gel electrophoresis**

66. The halogen iodine and chlorine inhabit ____function. ____**Protein**_____

67. Proteins from microbes are separated by _____. Electrophoresis

68. Endospores are stained with _____**Malachite green**

69. Vaccine of small pox _____ **Edward Jenifer**

70. _____alcohol is more effective than 100% _____**70%**

80. Hot air sterilized by _____ **Oxidation**

81. F means ____ **Fertility factor**

82. Bacteriophage is a _____**Virus**

83. The study of the evolutionary history of organism's _____ **Systematics/Phylogeny**

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84. Mycobiont is example of _____**Fungi**

85. Infectious circular RNA of plants_____**Viroid**

86. Consistently present, but with low incidence_____**Endemic**

1- The mode of action of chemicals to control micro-organism is the _____. **Phenol and phenolics**

2- Sight of entry through which micro-organism gain access to the body-_____ **Mucosal membrane**

3- Not true for virosoids_____. **dsDNA**

4- Fungi get CO2 from_____ **ORGANIC MATTER**

5- Prokaryotes are devided by_____ **MITOSIS**